

Hierarchical Regression on Housing Data


Group 6

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Project Objective

- Objective: Create a regression model that predicts the prices of houses in different cities and states based on house features
- Methods: Bayesian Hierarchical Regression

Data

- Data is from Kaggle
- Original data set contains 2.2 million houses from all 50 states
- Data contains information on house price, number of acres, number of square feet, number of bedrooms, number of bathrooms, state and city

Data Cleaning and Sampling

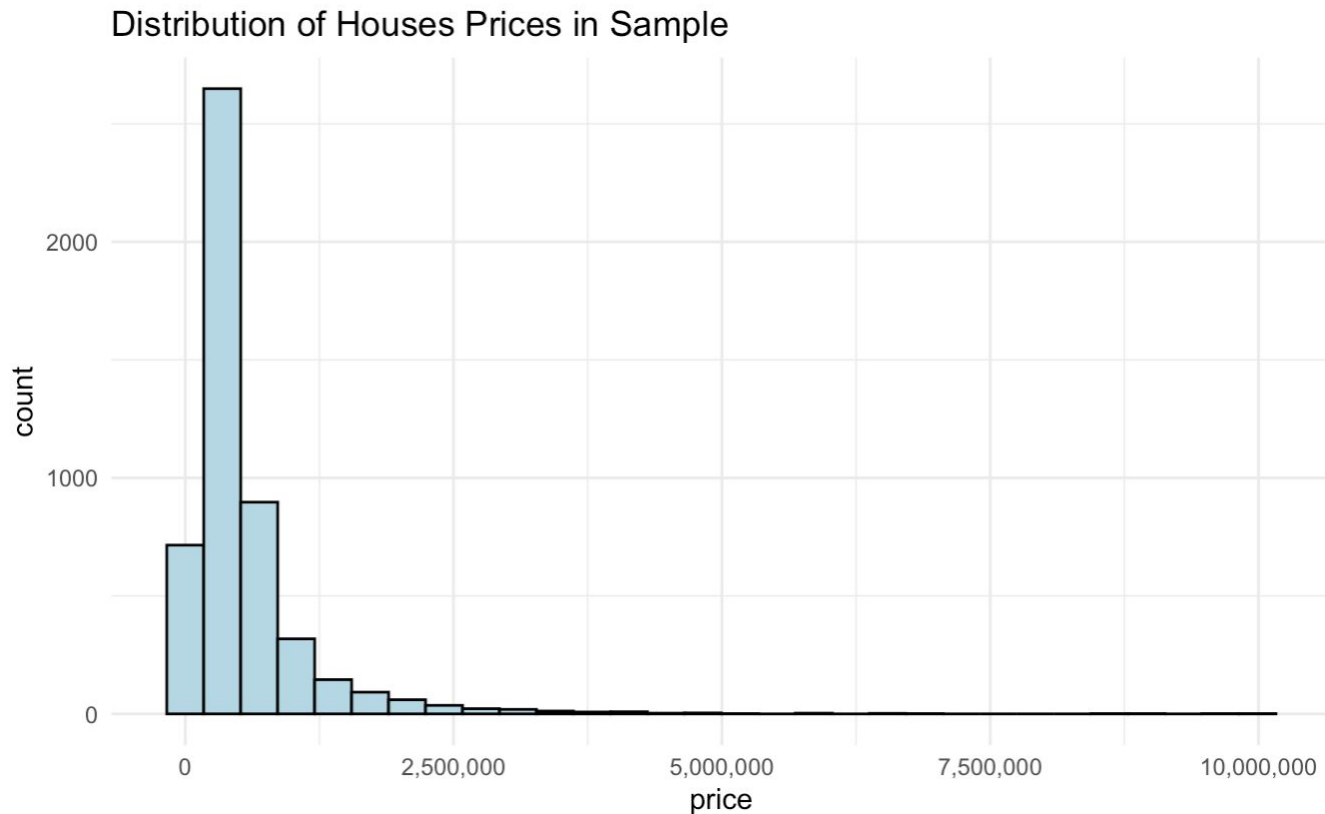
Data Cleaning:

- Removed rows with null values
- Removed outliers:
 - Houses over 10 million dollars
 - Houses over 50 acres
 - Houses over 10k square feet

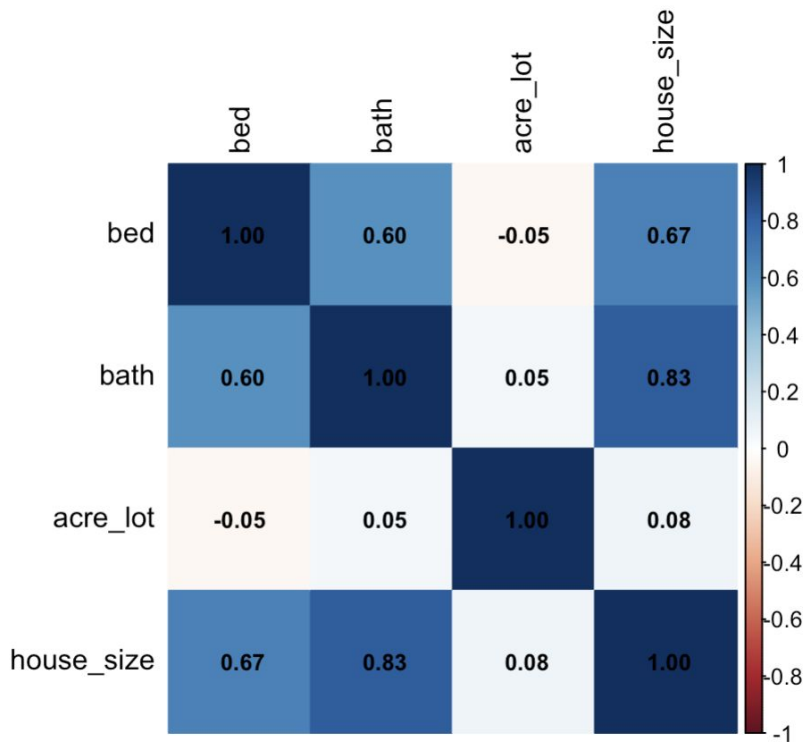
Data Sampling:

- Sampled from the following 5 states: Massachusetts, Michigan, California, Texas, and Florida
- For each state we sampled from 10 cities in that state
- Some cities included in the sample: Boston, Los Angeles, San Francisco, Ann Arbor, Detroit, Houston, and Miami
- For each city we sampled 100 houses
- The total sample contains 5,000 houses from 5 states and 50 cities

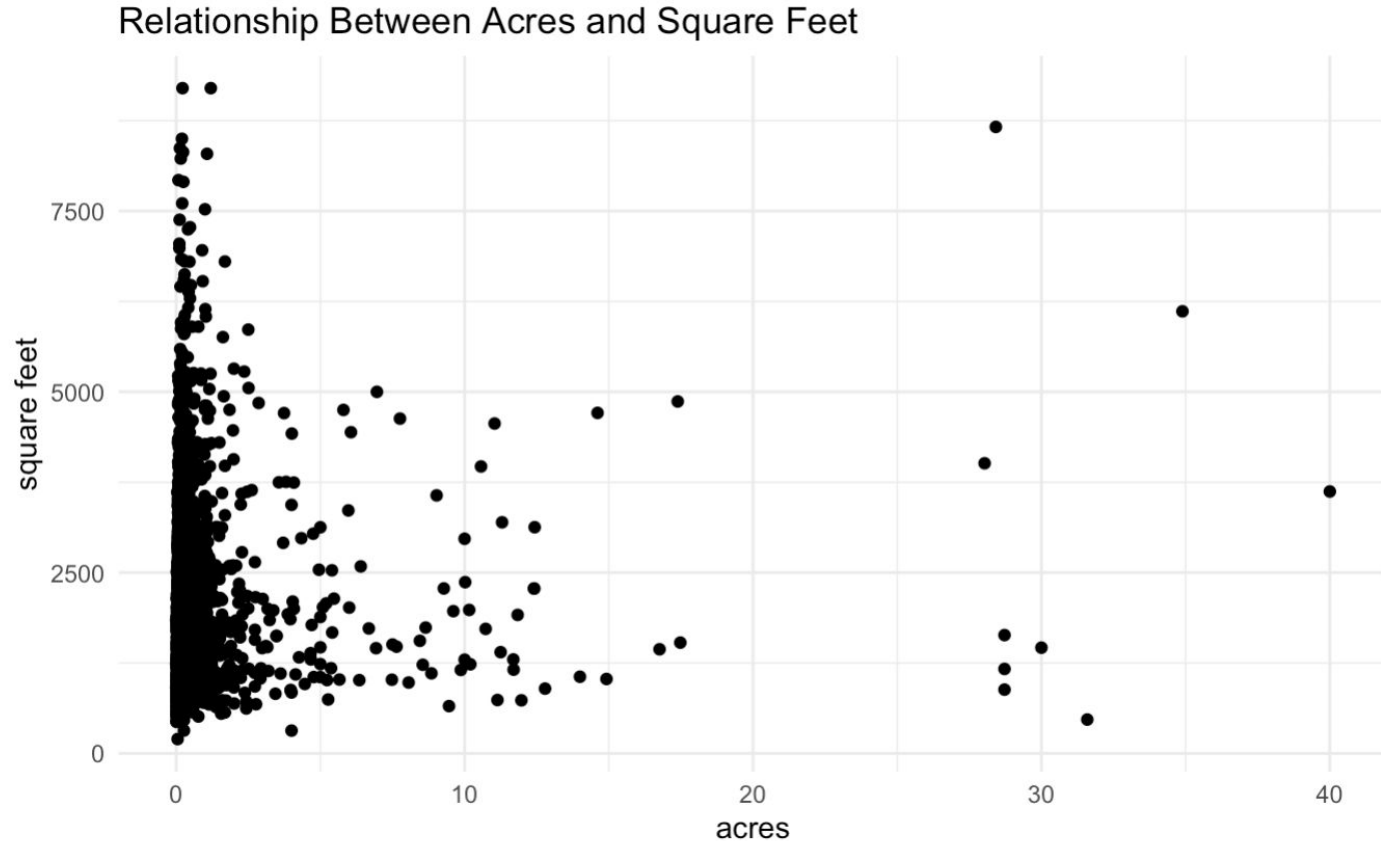
Data Visualization 1 - House Price Distribution



Data Visualization 2 - Correlation Matrix of Variables



Data Visualization 3 - Acres and Square Feet



Standardizing Predictors and Response Variable

Predictors:

- Acres
- Square Feet

Response:

- Price

Since the predictors and response variables are measured on different scales, we standardized the variables before running the regression analysis.

This makes the regression more stable, prevents numerical issues, and allows coefficients to be compared directly.

Baseline Regression Model

Simple Regression Model:

$$\widehat{\text{Price}} = \beta_0 + \beta_{\text{sq. feet}} * \text{sq. feet} + \beta_{\text{acres}} * \text{acres}$$

In-sample $R^2 = 0.226$

Not good at Predicting the House Prices

Bayesian Hierarchical Regression Model part 1

Let:

$s = 1, \dots, 5 \rightarrow$ states

$c = 1, \dots, 10 \rightarrow$ cities per state

$i = 1, \dots, 100 \rightarrow$ houses per city

$y_{i,s,c} \rightarrow$ house price

$x_{i,s,c} \rightarrow$ house covariates

Hyperprior:

$\omega \sim N(0, 2)$

with $\omega \geq 0$

Bayesian Hierarchical Regression Model part 2

State Level (Layer 1):

For each state $s = 1, \dots, 5$

$$\beta_s \mid \omega \sim N(\omega, \sigma_s^2)$$

$$\sigma_s^2 \sim N(0, 1)$$

$$\text{with } \sigma_s^2 \geq 0$$

~

City Level (Layer 2):

For each city $c = 1, \dots, 10$ in state s

$$\beta_{s,c} \mid \beta_s \sim N(\beta_s, \sigma_c^2)$$

$$\sigma_c^2 \sim N(0, 1)$$

$$\text{with } \sigma_c^2 \geq 0$$

Bayesian Hierarchical Regression Model part 3

Data Level (Layer 3):

For each house $i = 1, \dots, 100$ in city c in state s

$$y_{i,s,c} \mid \beta_{s,c} \sim N(\mu_{s,c}, \sigma^2)$$

$$\sigma^2 \sim N(0, 1)$$

$$\text{with } \sigma^2 \geq 0$$

Where:

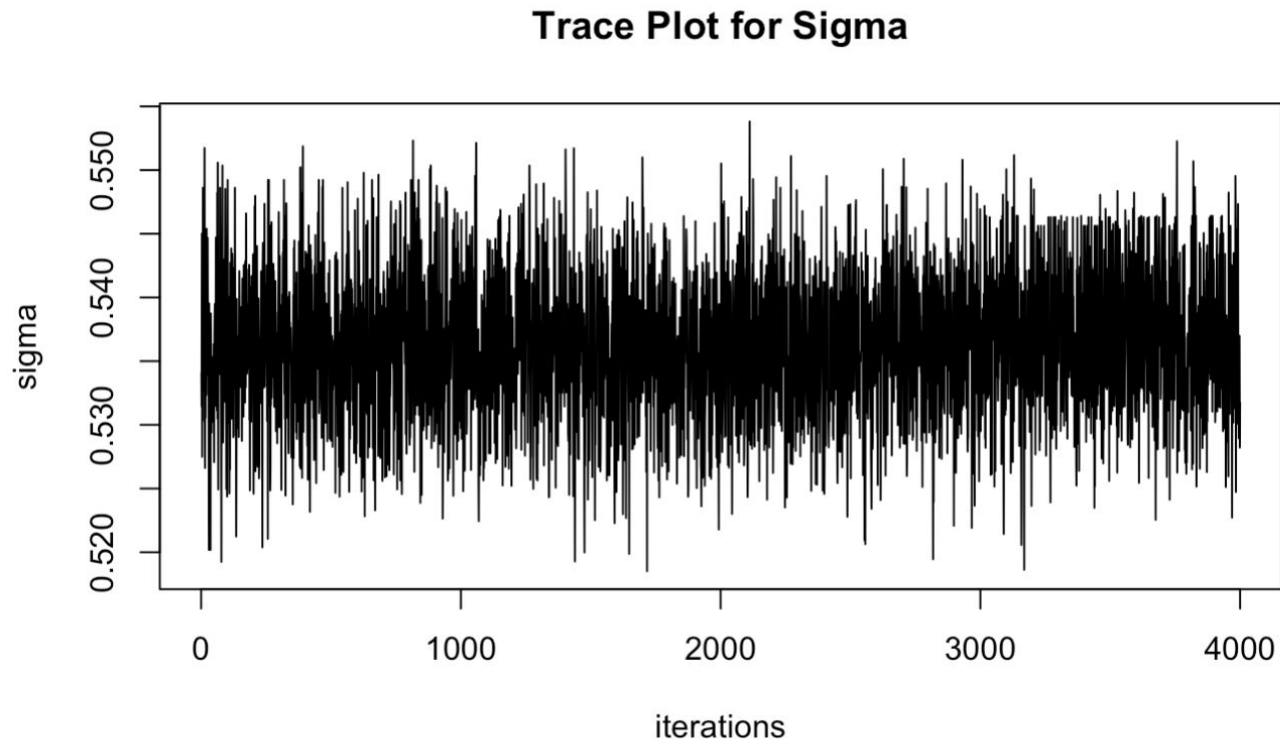
$$\mu_{s,c} = \beta_{s,c}^T x_{i,s,c}$$

Results for Hierarchical Regression Model

In-sample $R^2 = 0.72$

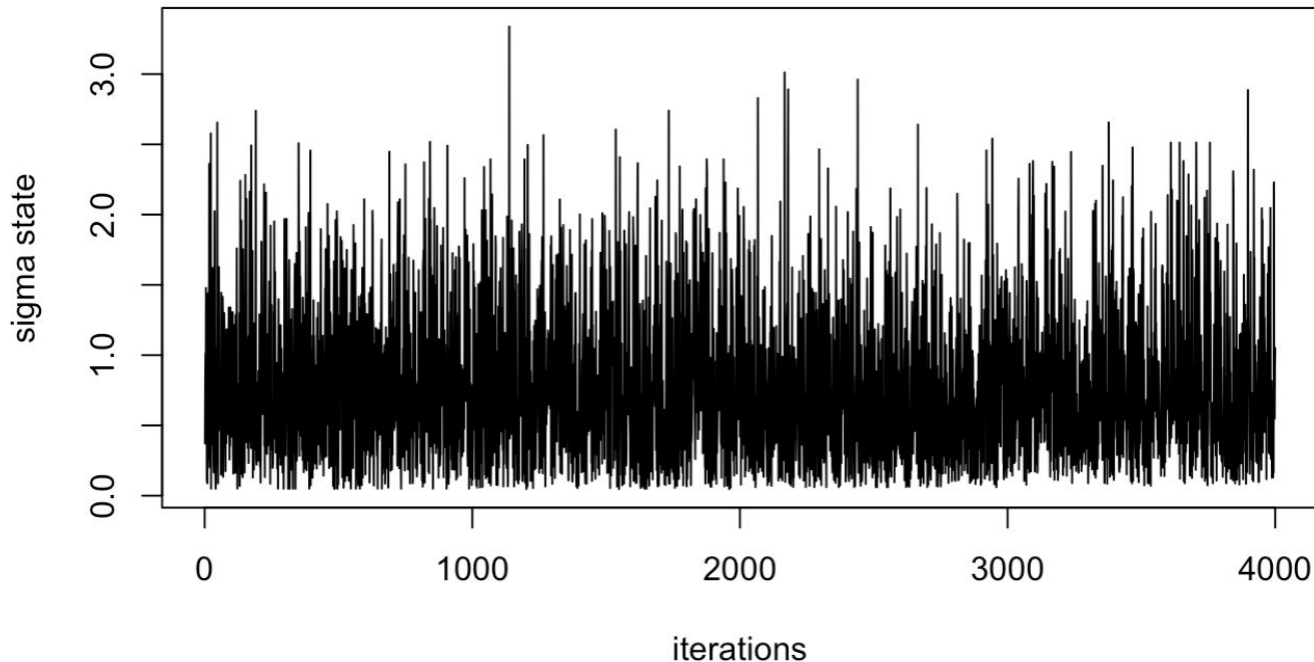
Much better than baseline regression model!

Markov Chains for Parameters - Sigma



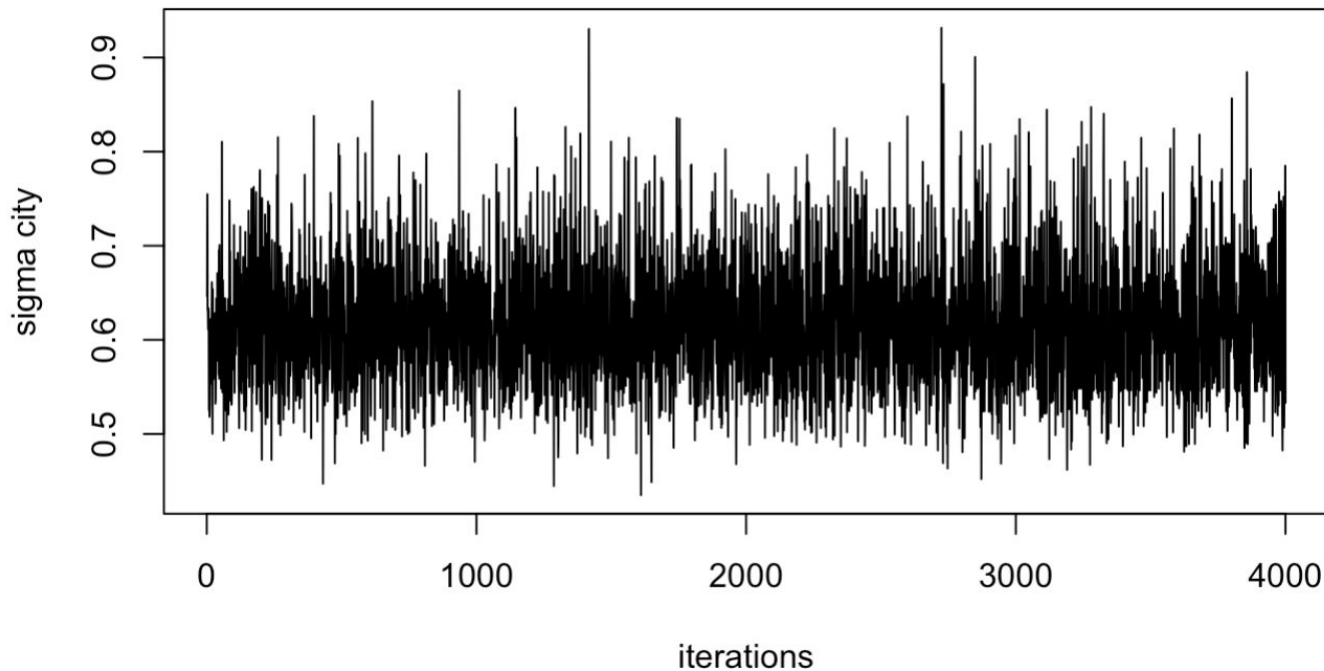
Markov Chains for Parameters - Sigma State

Trace Plot for Sigma State



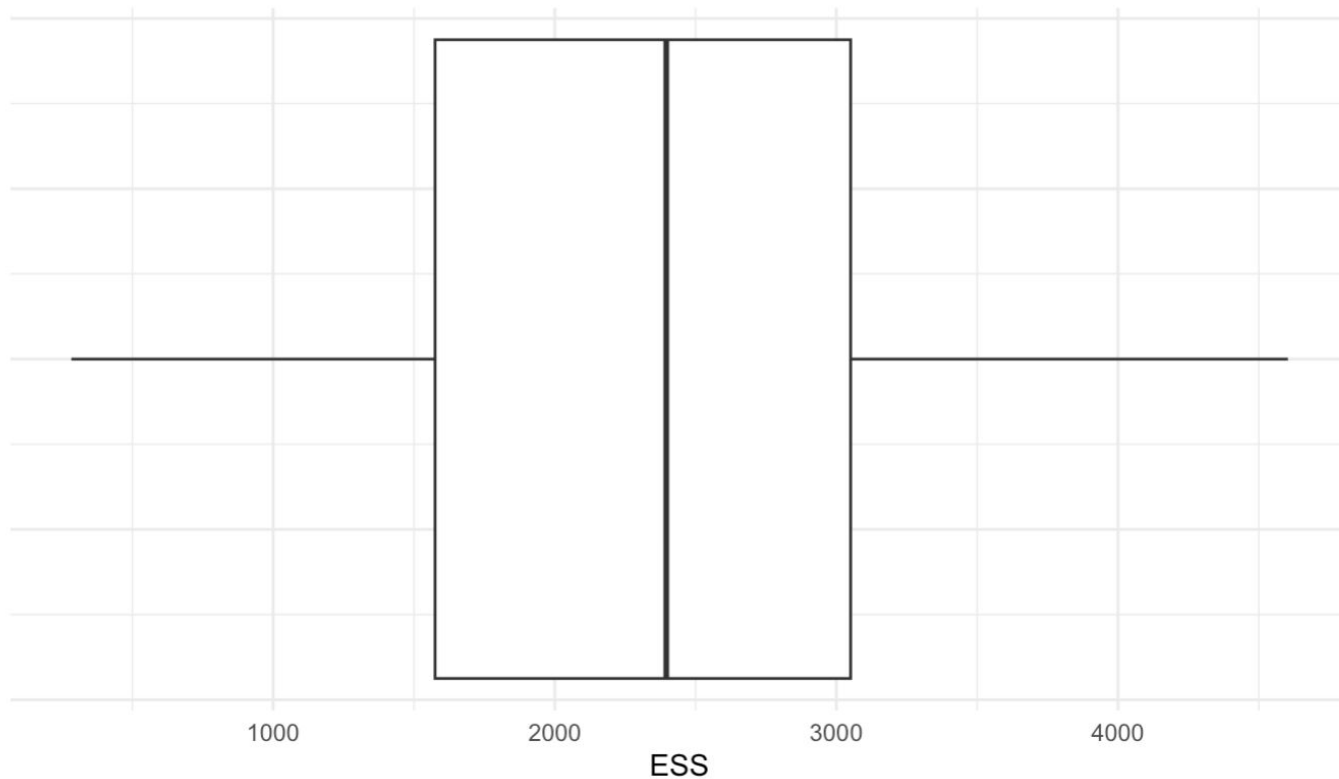
Markov Chains for Parameters - Sigma City

Trace Plot for Sigma City



Effective Sample Sizes for Parameters

Box Plot of ESS for all Parameters

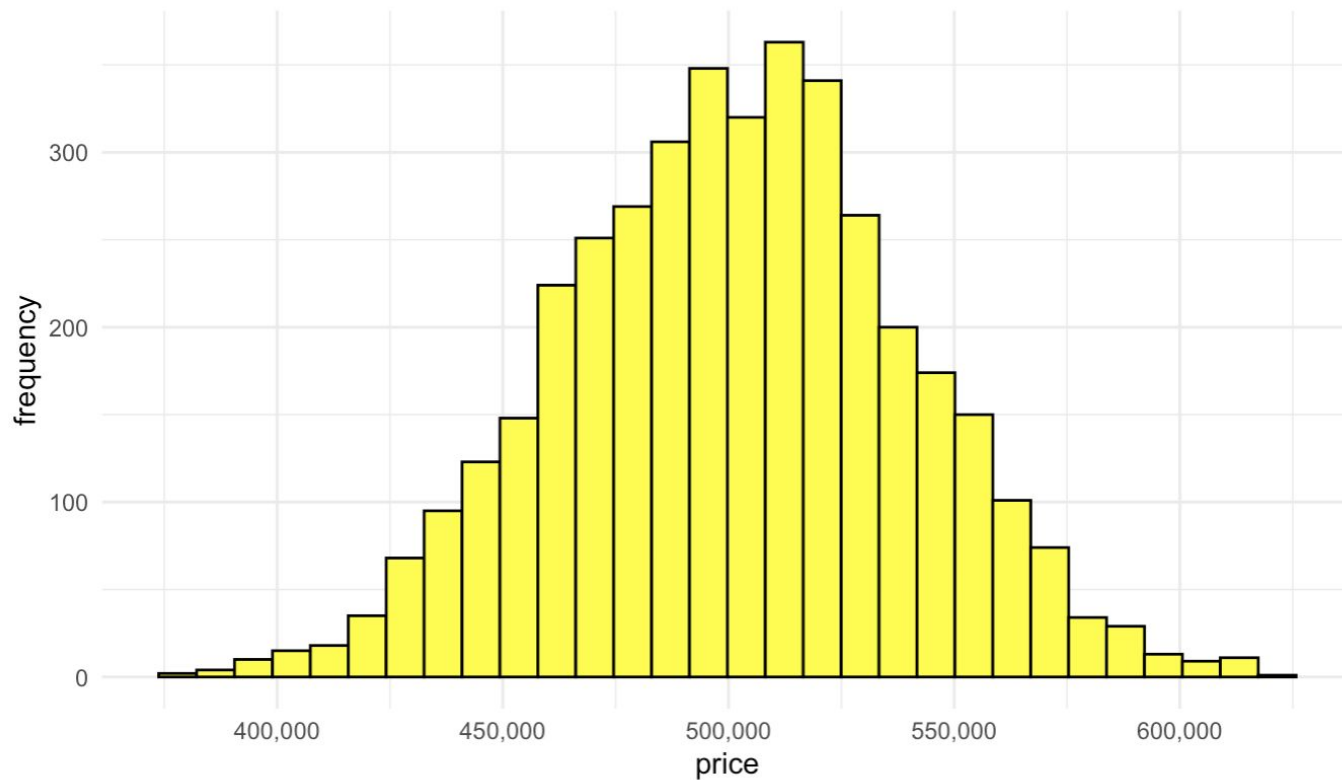


Posterior Price Predictions for a Given House in Different Cities

- For a given house with 0.2 acres and 1600 square feet
- We have the posterior price predictions from our bayesian hierarchical regression model

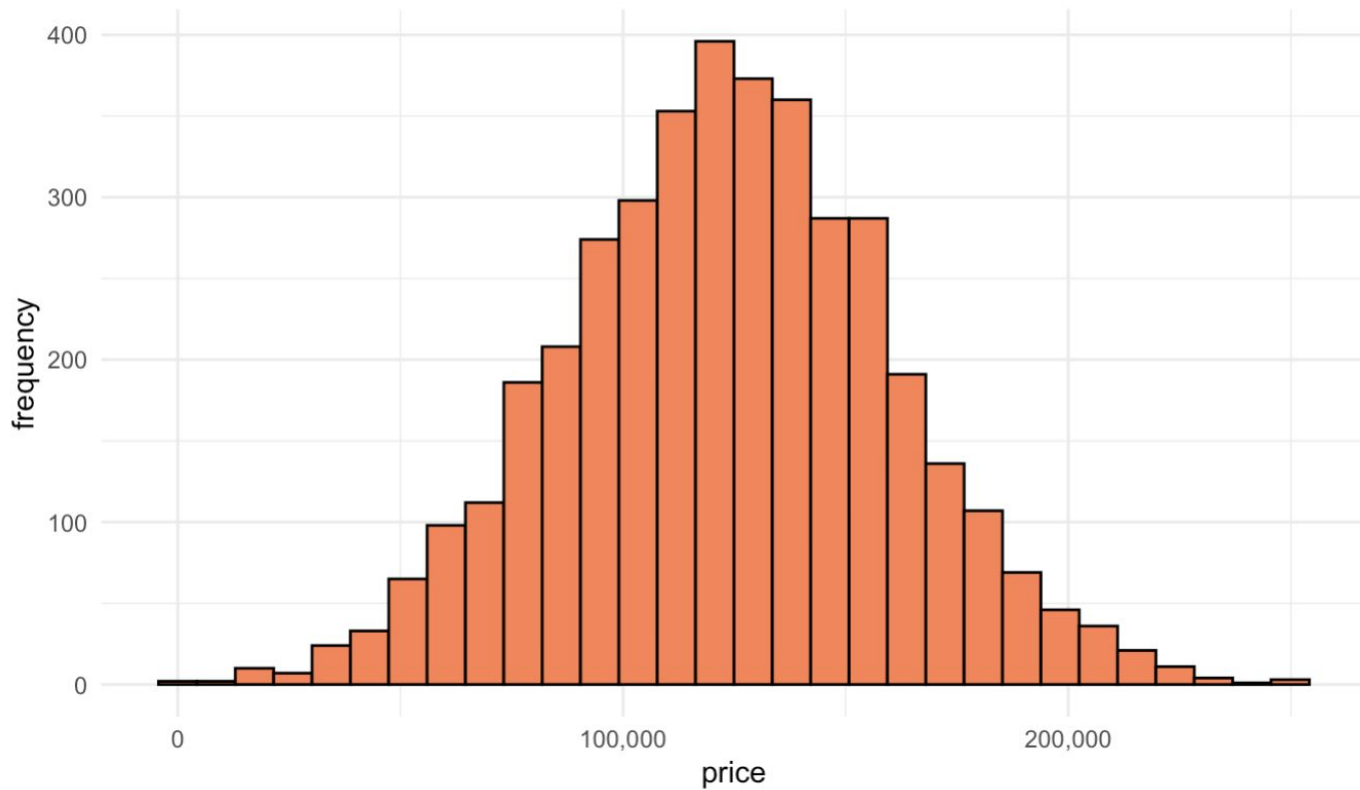
Ann Arbor

Posterior Price Prediction for Described House in Ann Arbor



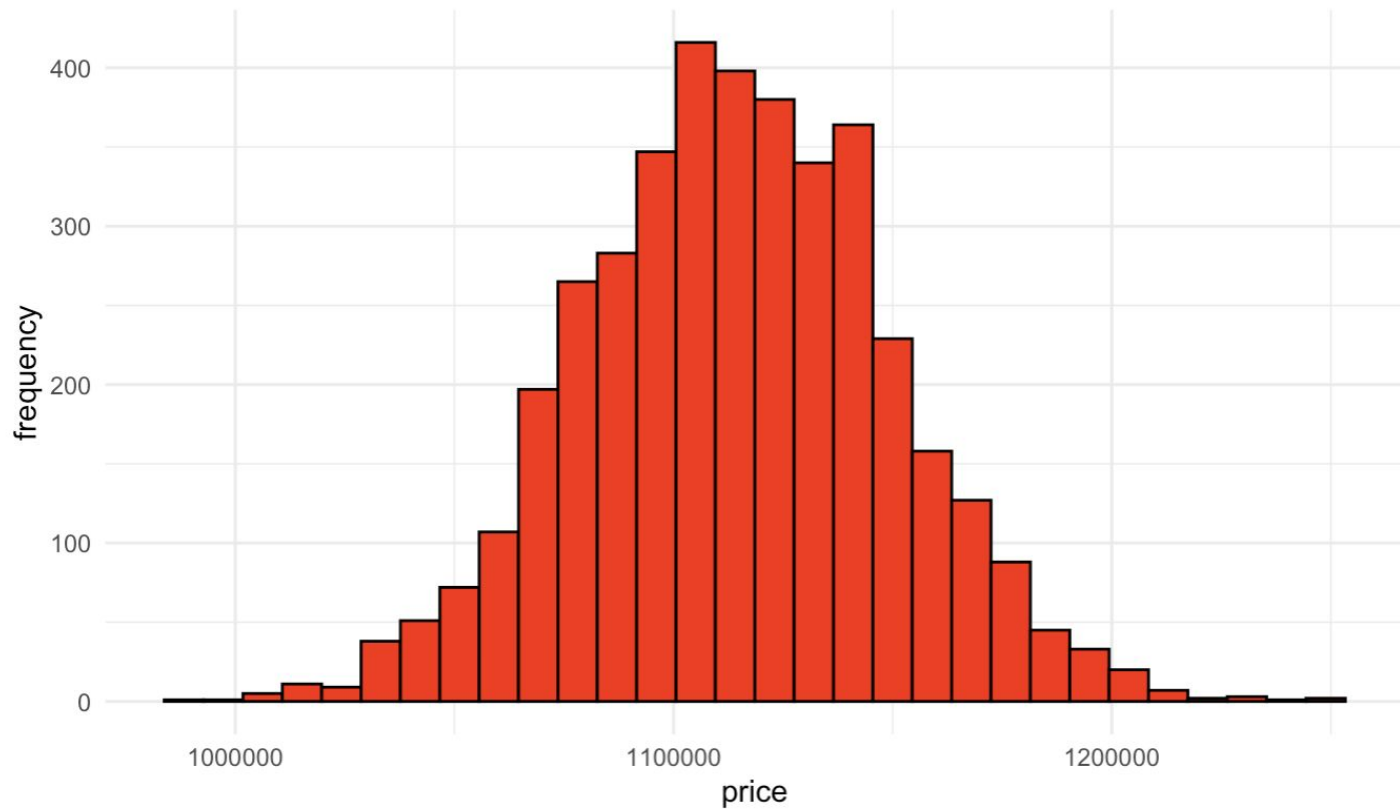
Detroit

Posterior Price Prediction for Described House in Detroit



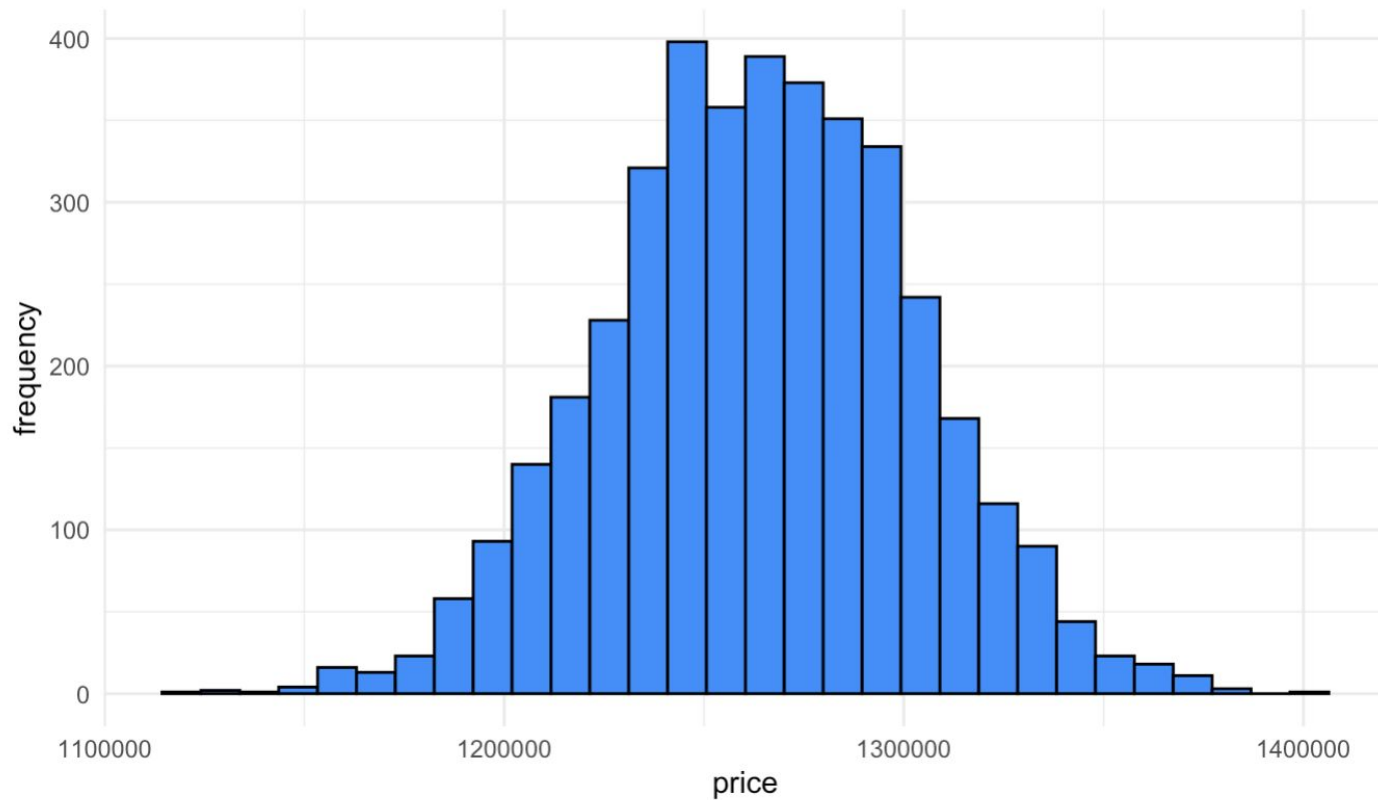
Boston

Posterior Price Prediction for Described House in Boston



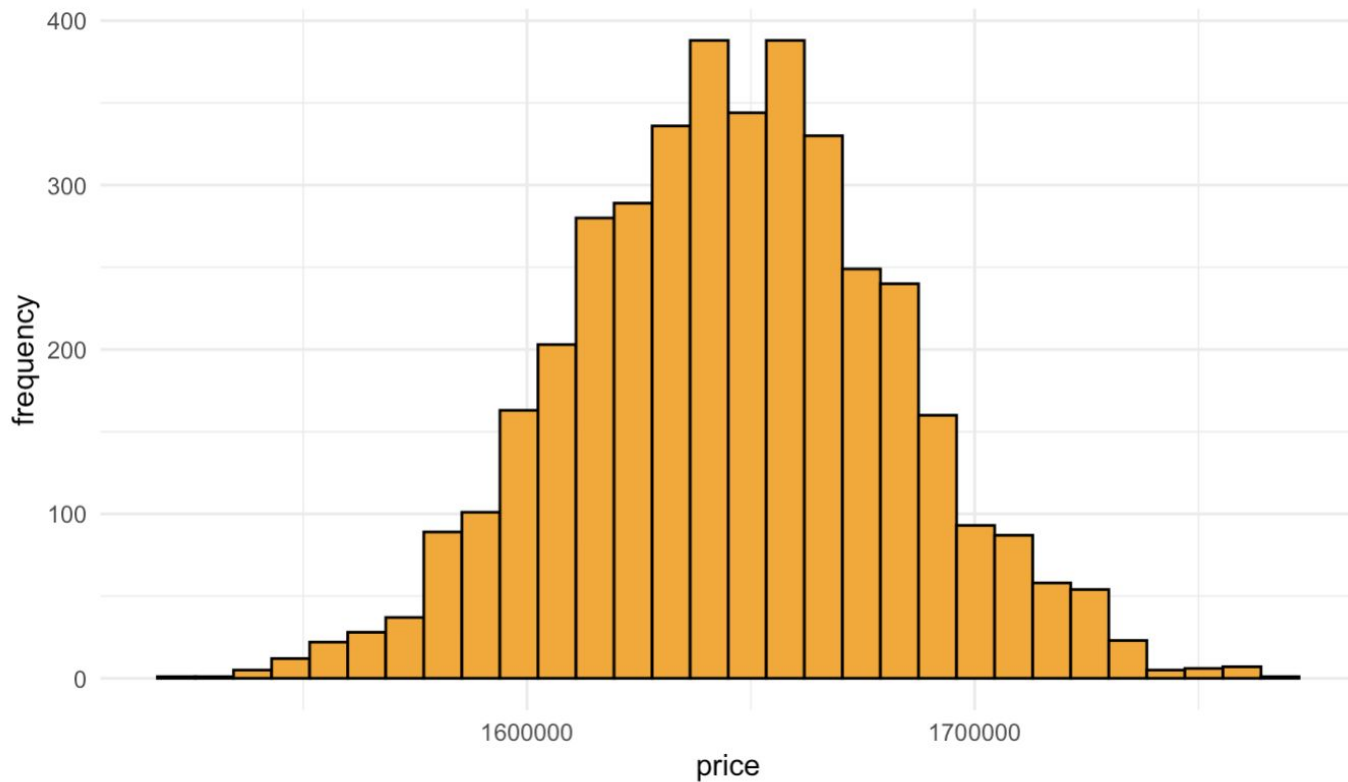
Los Angeles

Posterior Price Prediction for Described House in Los Angeles



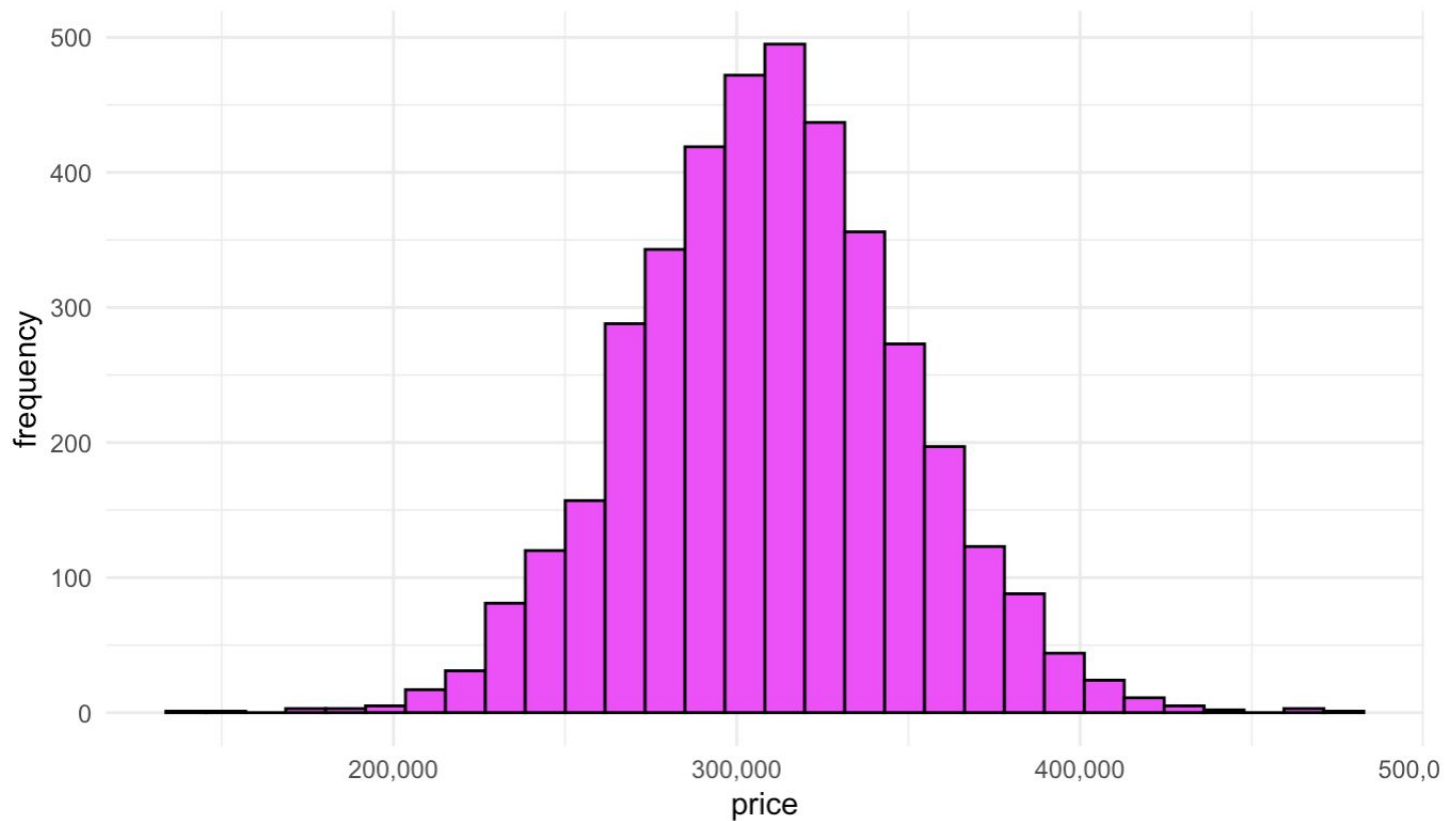
San Francisco

Posterior Price Prediction for Described House in San Francisco



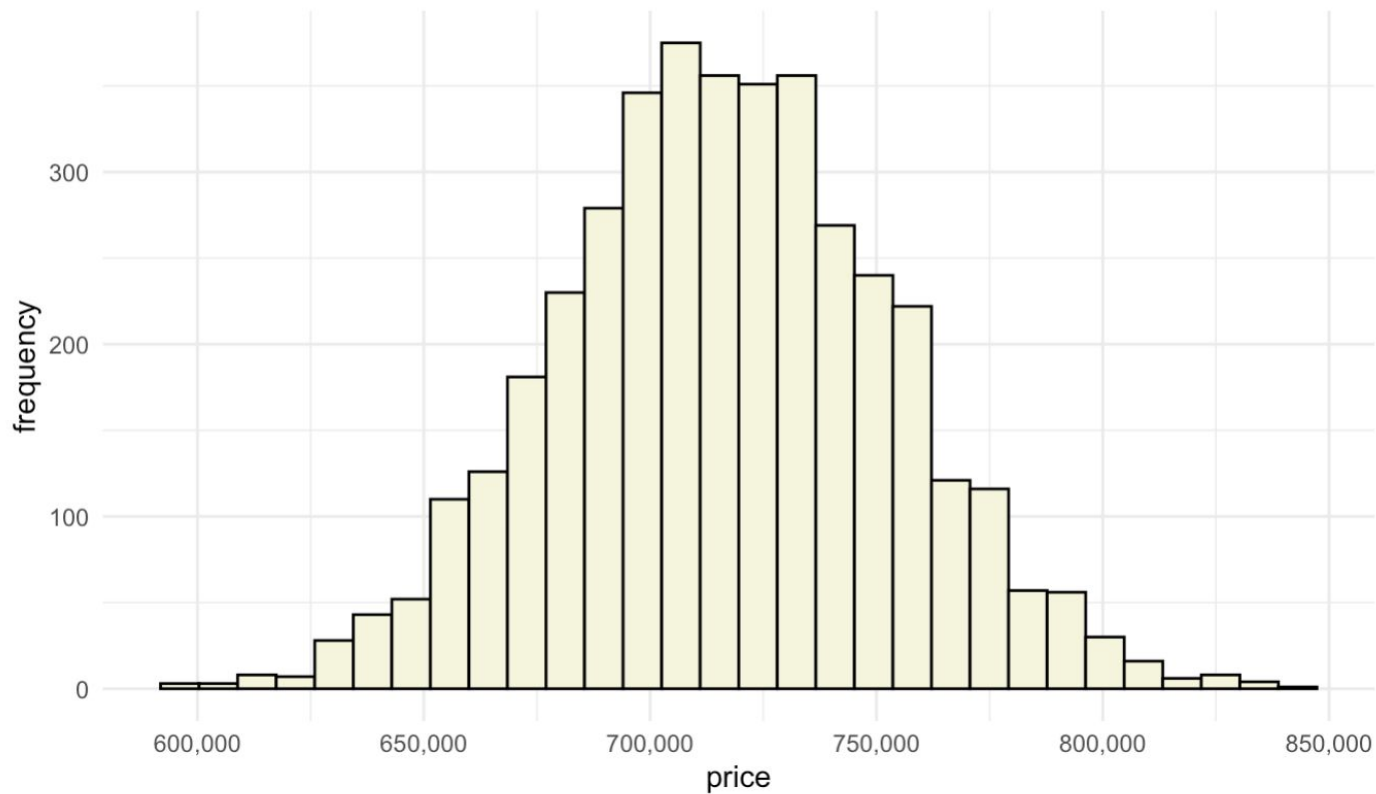
Houston

Posterior Price Prediction for Described House in Houston



Miami

Posterior Price Prediction for Described House in Miami



Sources

Sakib, A. S. (2024, March 30). *USA Real Estate Dataset*. Kaggle.

<https://www.kaggle.com/datasets/ahmedshahriarsakib/usa-real-estate-dataset/data>